P-S-600c
December 12, 1962

SUPERSEDING
Fed. Spec. P-S-600b
May 18, 1956

FEDERAL SPECIFICATION

SOAP, LAUNDRY, LOW-TITER (FOR LOW-TEMPERATURE WASHING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of laundry soap suitable for laundering and general cleaning purposes with soft water (see 6.1).

1.2. Classification.

1.2.1 *Grade, type, and forms.* The soap covered by this specification shall be of one grade and one type in the following forms, as specified (see 6.2):

Form 1—Chip or flake.

Form 2—Powdered or granular.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specification:

UU-S-48—Sacks, Shipping, Paper.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 536—Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Service Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-147 — Palletized Unit Loads.

FSC 7930

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(Copies of Military Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2. Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

American Society for Testing and Materials:

D 460-57T—Tentative Method of Test for Copper in Soap and Soap Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified in the contract or order (see 6.2) and before production begins, complete test results showing that the product meets the requirements in this specification and a 5-pound sample of the applicable end item shall be submitted to the contracting officer for approval. Approval of the preproduction sample authorizes production. Such authorization does not relieve the contractor of the responsibility for compliance with all provisions of this specification.

3.2 Composition. The composition of the soap shall conform to the requirements specified in table I, when tested as specified in table II. Deliveries containing a moisture and volatile matter content at 105°C. in excess of 7 percent shall be rejected without further tests.

3.2.1 *Computation of volatile matter.* The percentage of moisture and volatile matter at 105°C. shall be computed and reported by the testing laboratory on the soap as received. The percentage of all other constituents shall be calculated and reported on a moisture and volatile matter content of 7 percent. (See 6.4.) To adjust the percentage of the other constituents, where applicable, for samples having a moisture and volatile matter content at 105°C. of less than 7 percent, multiply the percentage of constituents requirement, other than moisture and volatile matter, by a factor (F) as follows:

$$F = \frac{100 - 7}{100 - \text{percent moisture and volatile matter (as received)}}$$

3.2.2 *Certificate of compliance.* A certificate of compliance shall be furnished by the supplier to the contracting officer stating

TABLE I.—*Composition*

	Minimum	Maximum
Moisture and volatile matter at 105°C., percent	....	7.0
Sum of free alkali or free acid, total matter insoluble in alcohol, and sodium chloride, percent	....	8.0
Free alkali, calculated as sodium hydroxide, (NaOH), percent	....	0.2
Free acid, calculated as oleic acid, percent	....	0.0
Matter insoluble in water, percent	....	1.5
Anhydrous soap, percent	86.0	....
Titer of the mixed fatty acids prepared from the soap, °C.	....	18.0
Rosin	....	0.0
Sugar	....	0.0
Water soluble sodium silicate, calculated as 1 Na_2O:3.25 SiO_2, percent	2.0	4.0
Copper, parts per million	....	10.0
Iodine number (Wijs) of the mixed fatty acids prepared from the soap	80.0	95.0
Organic oxidation inhibitor [1], percent	0.05	....
Unsaponified saponifiable matter, percent	....	1.0

[1] Acceptable inhibitors of the following types shall be used unless otherwise specified: diphenylamine; orthodiphenylbiguanidine; paratertiaryamylphenol; paratertiarybutylphenol. Contractor shall indicate which inhibitor will be used, and shall not change inhibitor during production unless approved by the contracting officer.

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that the organic oxidation inhibitor content complies with the requirements in table I. The contractor shall state which inhibitor has been used, and shall not change inhibitor during production without approval of the contracting officer.

3.3 Color. The soap shall have a uniform light color. Unless otherwise specified, the color of a freshly sampled soap shall be not darker than color number 30266 of Fed. Std. No. 595 (see 6.3).

3.4 Odor. The soap shall be free of an objectionable odor as received, and in a soap solution in water at 125°-130°F., when tested at specified in table II. An objectionable odor, in this specification, is defined as a rancid, fishy, or horsey odor.

3.5 Stability. The soap shall be resistant to the development of rancidity. The product shall not develop an objectionable odor when tested as specified in table II.

3.6 Form.

3.6.1 *Form* 1. The soap shall be in chip or flake form and no particle shall exceed 3 grams in weight when tested as specified in table II.

3.6.2 *Form* 2. The soap shall be in powdered or granular form and a maximum residue of 1.5 percent shall be retained on a number 12 sieve when tested as specified in table II.

3.7 Workmanship. The finished product shall be clean, uniform, free flowing, and shall not lump or cake.

3.8 Quality. The quality of the finished product shall conform to the levels established herein. The occurrence of defects shall not exceed the number permitted under the acceptable quality levels (AQLs) specified in section 4 for any examination and tests of materials, components and end items, as applicable.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. The supplier is responsible for the performance of all the inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Sampling for inspection and tests.* Sampling for inspection and tests shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, a lot shall consist of all end items of the same form, made from the same components and submitted for delivery at one time.

4.2 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that all the materials and components used in the end item were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined in accordance with the classification of defects set forth in the following sub-paragraphs at the inspection levels and acceptable quality levels (AQLs) set forth in paragraph 4.3.5. Ran-

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dom samples shall be drawn from each lot of end items. The lot size for purposes of determining the sample size, in accordance with MIL-STD-105, shall be expressed in units of containers filled with the same form product, consisting of soap manufactured in one batch using the same materials and offered for delivery at one time for the examinations in 4.3.2, 4.3.3, and 4.3.4.

4.3.2 *Examination of the soap product for defects in form, workmanship, color, and odor.* The sample unit for this examination shall be the contents of one filled container.

Examine	*Defect*
Form:	
1	Not chips or flakes.
2	Not powdered or granular.
Workmanship	Not clean; not uniform; any foreign matter.
	Not free flowing.
	Lumping or caking.
Color	Not uniform.
	Too dark; darker than color chip number 30266, unless otherwise specified.
Odor	Any objectionable odor (rancid, fishy, or horsey odor).

4.3.3 *Examination of the end item for defects in net contents.* The sample unit for this examination shall be one filled container. The average net contents per container shall be not less than the specified or indicated quantity.

4.3.4 *Examination of preparation for delivery.* An examination shall be made to determine that construction and markings comply with the packing requirements of section 5 of this specification. The sample unit shall be one shipping container fully prepared for delivery.

Examine	*Defect*
Construction	Not in accordance with contract requirements.
	Container not as specified.
	Closures or seams not accomplished by specified or required methods or materials.
	Tuck-in sleeves not moisture proof.
	Any seepage of contents.
	Any split, break, dent, hole, tear, puncture, or other defect which May affect serviceability.
	Thickness of filled sack not uniform when laid flat.
	Flat height more than 30 percent of the width of the container.
	Not palletized, when specified.
Markings	Omitted; illegible; incomplete; smeared; incorrect.
	Not marked "STORE IN A COOL, DRY PLACE".
	Not as specified.

4.3.5 *Inspection levels and acceptable quality levels* (AQLs) *for examinations.* The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per one hundred units shall be as follows:

4.3.6 *Testing of the soap product.* The soap shall be tested for characteristics specified in table I in accordance with test methods specified in table II for each lot presented for examination. The sample unit, for purposes of testing, shall be a one pound composite sample selected in accordance with 4.3.7. Unless otherwise specified, two determinations shall be made on the composite sample. The average of the two determinations shall comply with the requirements when numerical results are specified. Both determinations shall pass when "pass" or "fail" requirements are applicable.

Examination paragraph	*Inspection levels*	*AQLs*
4.3.2	L-4	4.0
4.3.3	L-3	..
4.3.4	I-S	2.5

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TABLE II.—*Instructions for testing*

Characteristic	Specification reference		Number determinations per composite sample unit	Results reported as	
	Requirement	Test method		Pass or fail	Numerically to nearest[3]
Moisture and volatile matter at 105°C. [1,2]	Table I	201.1 302.3 301.3	Average of 2	..	0.1 percent
Sum of free alkali or free acid, total matter insoluble in alcohol, and sodium chloride (NaCl) [1]	Table I	501	Average of 2	..	0.1 percent
Free alkali, calculated as sodium hydroxide (NaOH) [1]	Table I	302.3	Average of 2	..	0.1 percent
Free acid, calculated as oleic acid [1]	Table I	302.3	Average of 2	..	0.1 percent
Matter insoluble in water [1]	Table I	303.1	Average of 2	..	0.1 percent
Anhydrous soap [1]	Table I	401.3	Average of 2	..	0.1 percent
Titer of the mixed fatty acids prepared from the soap [1]	Table I	901	Average of 2	..	0.1°C.
Rosin [1]	Table I	701.3; Test No. 2	1	.. X	
Sugar [1]	Table I	1701	1	X	
Particle weight (form 1 only)	3.6.1	4.4.5	1	X	
Residue retained on a number 12 sieve (form 2 only) [1]	3.6.2	2101	Average of 2	..	0.1 percent
Water soluble sodium silicate, calculated as 1 Na_2O:3.25 SiO_2 [1]	Table I	1301	Average of 2	..	0.1 percent
Copper, parts per million	Table I	ASTM-D460-57T	2	..	
Iodine number (Wijs) of the mixed fatty acids prepared from the soap [1]	Table I	1101.2	Average of 2	X	number
Unsaponified saponifible matter [1]	Table I	603.1	Average of 2	..	0.1 percent
Odor	3.4	4.4.3	1	X	
Stability test	3.5	4.4.4	1	X	

[1] Test is performed in accordance with Fed. Test Method Std. No. 536.
[2] Upon failure of a sample by method 201.1 the results shall be verified by method 202.2.
[3] Test reports shall include all values on which results are based.

4.3.7 *Composite sample.* When four or more containers comprise the shipping lot, a sample of not less than 1 pound shall be taken from each of not less than four separate containers, randomly selected throughout the lot, presented for examination. When less than four shipping containers comprise the lot, the total samples shall, in all cases, consist of not less than 3 sample portions taken at random from the containers or container. The samples shall be placed in a receptacle of suitable size. After mixing thoroughly, a one 1 pound composite sample shall be drawn for testing purposes. The composite sample shall be placed in an air-tight container, sealed and marked with the identifying lot and date.

4.4 Test procedures.

4.4.1 *Composition.* The composition shall be determined according to the methods specified in Fed. Test Method Std. No. 536 as indicated in table II.

4.4.2 *Copper analysis.* Copper shall be determined in accordance with method D460-57T of American Society for Testing Materials.

4.4.3 *Test for odor of soap solution.* A one-ounce-per-gallon of distilled or demineralized water soap solution shall be prepared and heated to 125°-130°F. The soap solution shall be maintained at this temper-

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ature for 15 minutes. Determine the odor of the soap solution.

4.4.4 *Stability test.* A quart size glass fruit jar shall be half filled with the product. A 15 by 175 millimeter test tube, two-thirds full of water, shall be placed in the center of the soap product so that the bottom of the test tube rests on the bottom of the jar and so that the water vapor may freely escape into the jar. The soap product shall be lightly packed by gently tapping on the table top. Soap shall be added during tapping so that the jar remains half filled. The jar shall be sealed with a suitable gasket, making sure that no metal is exposed within the jar. The jar shall be placed in an oven or cabinet and maintained at 125° to 130°F. for a period of 15 days. At the end of 15 days, the cover shall be removed and the odor determined.

4.4.5 *Particle weight (form I).* Ten of the largest observed particles shall be selected from the composite sample. Each particle shall be weighed separately. The weight of each particle shall not exceed 3 grams.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packing for civil agency procurement.)

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 *Level A.* Fifty pounds of the soap shall be packed in a paper shipping sack conforming to type I, II, or IV, construction number 13X, with dipped ends, of UU-S-48. Type IV sacks shall be constructed with moisture-proof tuck-in sleeves. Closure of sacks shall not be completed until the soap contents have cooled to room temperature. The thickness of the filled sack, when laid flat, shall have a uniform thickness of not more than thirty percent of its width in that position. When specified, sacks shall be palletized in accordance with MIL-STD-147 (see 6.2).

5.1.2 *Level B.* Fifty pounds of the soap shall be packed in a paper shipping sack conforming to type I, II, or IV, construction number 38 L/W of UU-S-48. Closure of sacks shall not be completed until the soap contents have cooled to room temperature. The thickness of the filled sack, when laid flat, shall have a uniform thickness of not more than thirty percent of its width in that position. When specified, sacks shall be palletized in accordance with MIL-STD-147 (see 6.2).

5.1.3 *Level C.* Soap shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping sacks shall be marked in accordance with Fed. Std. No. 123. Each shipping sack shall be marked with the following precautionary marking: "STORE IN A COOL, DRY PLACE".

5.2.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping sacks shall be marked in accordance with MIL-STD-129. Each shipping sack shall be marked with the following precautionary marking: "STORE IN A COOL, DRY PLACE".

6. NOTES

6.1 Intended use. This specification covers a neutral laundry soap, substantially free from uncombined alkali or fatty matter. The soap is intended primarily for use with soft water in cleaning wool, synthetic fabrics, and finished surfaces where low tem-

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perature (90°-100°F.) washing solutions are indicated.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Form required.

(c) Whether or not a preproduction sample is required (see 3.1).

(d) Shade of color, when required (see (3.3).

(e) Inspection responsibility, methods, and test (see section 4).

(f) Selection of applicable level of packing (see 5.1).

(g) Whether shipping sacks shall require palletizing (see 5.1.1 and 5.1.2).

6.3 Color chips. Color chips listed in Fed Std. No. 595 may be purchased at five cents each from any Business Service Center of General Services Administration.

6.4 Basis for purchase. The soap shall be purchased by new weight provided the matter volatile at 105°C is neither above or below 5 percent. Deliveries containing more than 7 percent of matter volatile shall be rejected without further test. On deliveries containing 7 percent or less of matter volatile at 105°C payment should be made on a basis of a product containing 5 percent moisture. The net weight basis is calculated as follows:

$$W = \frac{R \times (100\text{-}L)}{95}$$

Where: W = net weight to be paid for on 5 percent moisture and matter volatile basis.

R = net weight of material as received.

L = percentage loss at 105° C.

6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Soap, not otherwise indexed by name, other than liquid.

Carload minimum weight 36,000 pounds.

Motor:

Soap, not otherwise indexed, other than liquid.

Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY CUSTODIANS:

Army—Q

Navy—Sh

Air Force—MOA

OTHER INTEREST:

Army—M

Navy—Wep Y

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